

NETWORKS AND COMPLEXITY

Solution 16-2

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 16.2: Fold bifurcation normal form [Lvl. 2]

Compute the steady states and their stability for the system

$$\dot{x} = ax^2 + bp.$$

Solution

We compute the steady states by solving

$$0 = ax^2 + bp \tag{1}$$

which yields

$$x^* = \pm \sqrt{-\frac{bp}{a}} \tag{2}$$

This means we have either two steady states ($bp/a < 0$) or none ($bp/a > 0$). If a and b have the same sign, then steady states exist at negative values of p , otherwise they only exist at positive values of p .

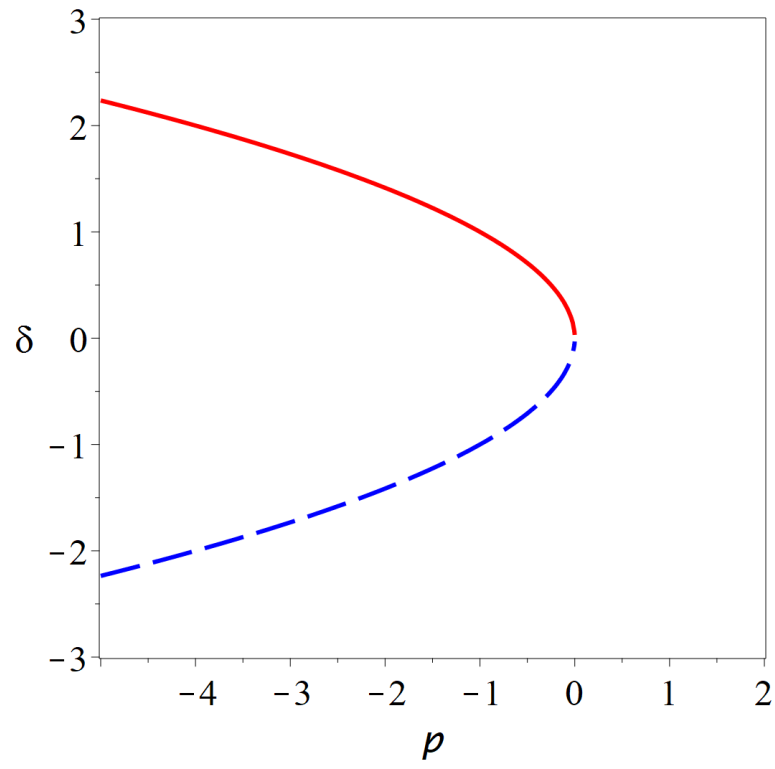
In this sense there are two forms of the fold bifurcation, but they are indistinguishable in practice.

To determine the stability of solutions we compute

$$f_x = \frac{\partial}{\partial x} ax^2 + bp = 2ax \tag{3}$$

If $a > 0$ then the positive steady state is unstable and the negative one is stable. For $a < 0$ it's the other way around.

We can now draw the bifurcation diagram



This diagram corresponds to the case $a < 0$ (positive branch of steady states is stable) and $b < 0$ (steady states exist for negative p).